SUSTAINABILITY

Performance-Normalized Embodied Rack Carbon Intensity for Next-Generation Hardware

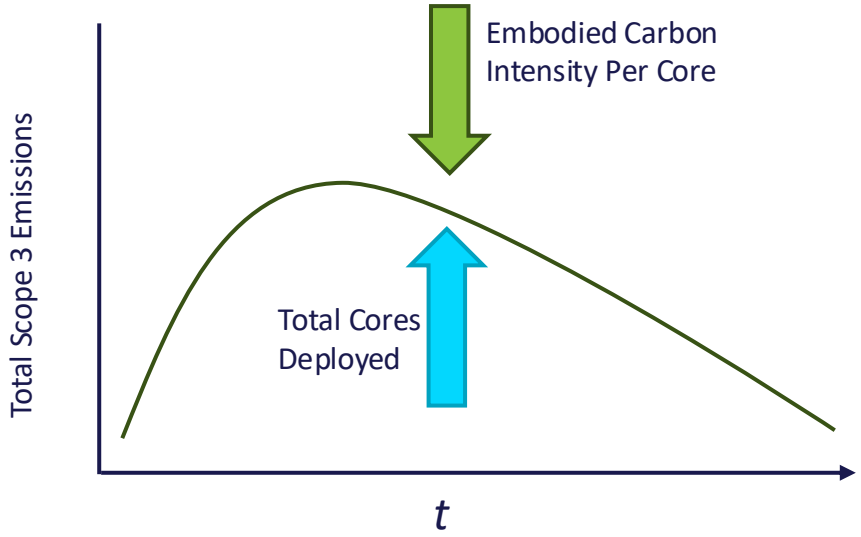
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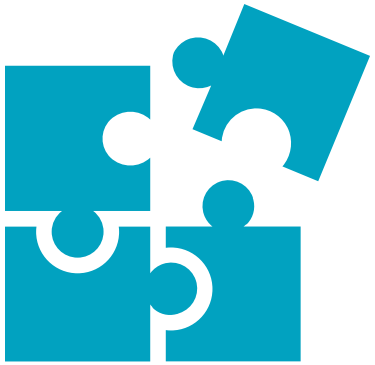
OCP PLATINUM Member

Why intensity metrics matter in the AI growth era

- Rapid growth in Cloud is leading to an increase in near-term absolute emissions even as IT hardware becomes more efficient and meaningful reduction interventions are in place
- To track and communicate progress, we need complementary metrics that track performance-based intensity (per part or per rack) *alongside* absolute emissions



Performance-normalized embodied rack carbon intensity



- Holistic environmental metrics should account for emissions across the entire stack: Hardware + Software + Infrastructure[1,2] and is required for comprehensive attributional accounting[3]
  - However, there can be benefits (*and tradeoffs*) to focusing on 1 piece of the puzzle
- Focus on IT Hardware embodied carbon intensity
  - Purpose - Isolate IT hardware supply chain decarbonization from larger macrotrends in HW and SW efficiency
  - Remove the complexity and challenges of calculating a comprehensive metric to help focus attention
  - Use simple, business-aligned functional units for intensity that can be translated from parts (suppliers) to HW systems.

1. Wang etl al, 2025. CATransformers: Carbon Aware Transformers Through Joint Model-Hardware Optimization
2. Schneider et al, 2025. Life Cycle Emissions of AI Hardware: A Cradle-To-Grave Approach and Generational Trends Estimating and Reporting
3. Russell, Stephen, 2019. The Comparative Emissions Impacts of Products

Measuring Carbon Intensity of Storage, Compute & AI

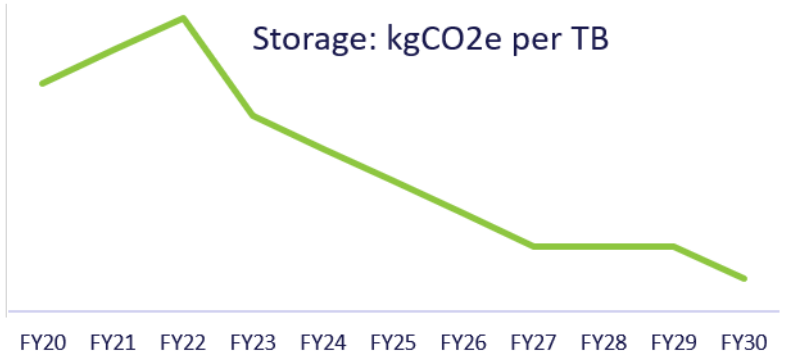
- Rack-level embodied carbon intensity is useful to track emissions progress across compute, storage, and AI hardware
- Emissions are normalized by common units of performance such as Compute Cores, Storage TBs, and GPU FP8TOPs or TFLOPs.
- Tracked over time to show portfolio level progress in Scope 3 by major HW types
  - Uneven progress due to dynamic changes in semiconductor technology, product mix, architectural evolution, etc.

$$Storage_{Annual} = \frac{Total\ Annual\ Storage\ Emissions\ (kgCO2e)}{Total\ Annual\ Storage\ TBs}$$

$$Compute_{Annual} = \frac{Total\ Annual\ Compute\ Emissions\ (kgCO2e)}{Total\ Annual\ Compute\ Cores}$$

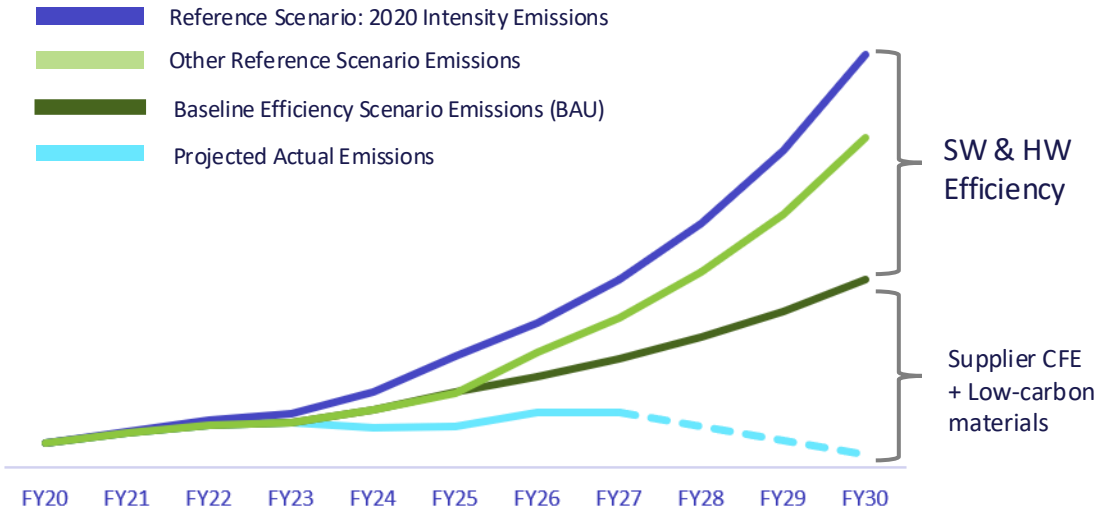
$$GPU_{Annual} = \frac{Total\ Annual\ GPU\ Emissions\ (kgCO2e)}{Total\ Annual\ GPU\ FP8TOPs}$$

Hypothetical Storage Intensity over time



Constructing Emissions Scenarios

- **Reference Scenario: 2020 Intensity** – Uses a reference year emissions intensity multiplied by current year procurement to demonstrate an emissions counterfactual (*i.e. avoided emissions*).
  - **Other Reference Scenarios** – Opportunity to further decouple the impact of SW & HW efficiency (e.g. semiconductor tech evolution vs CSP delivery innovations)
- **Baseline Efficiency Scenario** – Emissions that demonstrate hardware and software efficiency improvements that are part of normal operations but do not include supply chain interventions
- **Actual Emissions** – Baseline emissions with realized interventions



Hypothetical emission scenarios

Sample Calculations

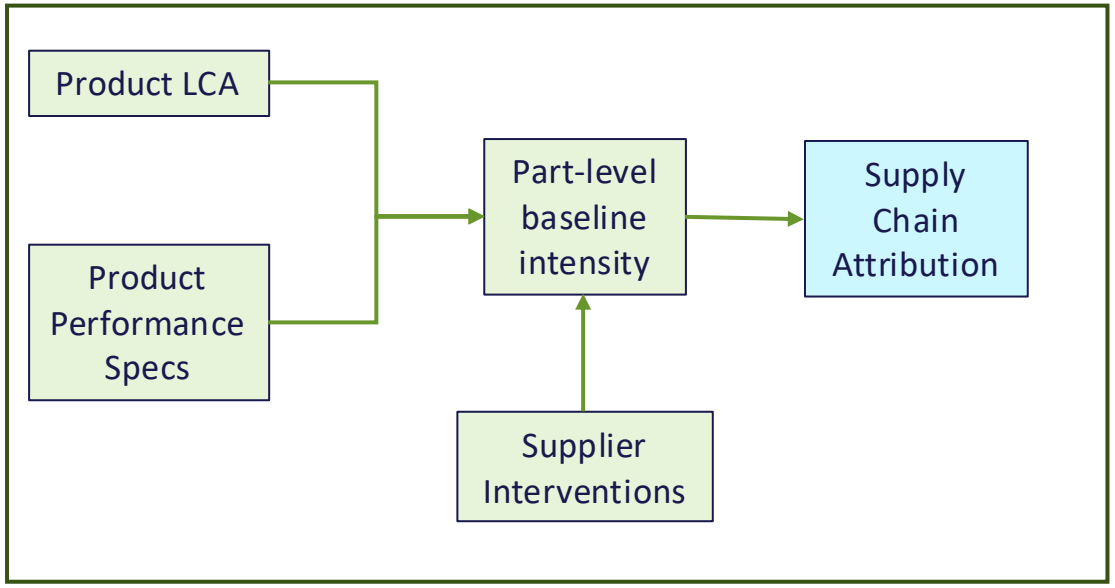
$$\boldsymbol{Reference\ Emissions_{FY30}} = Total\ TBs_{FY30} * \frac{kgCO2e}{TB}_{FY20} + Total\ Cores_{FY30} * \frac{kgCO2e}{vCore}_{FY20} + Total\ FP8TOPs_{FY30} * \frac{kgCO2e}{FP8TOP\ or\ TFLOPs_{FY20}}$$

$$\boldsymbol{Baseline\ Emissions_{FY30}} = Total\ TBs_{FY30} * \frac{kgCO2e}{TB}_{FY30} + Total\ Cores_{FY30} * \frac{kgCO2e}{vCore}_{FY30} + Total\ FP8TOPs_{FY30} * \frac{kgCO2e}{FP8TOP\ or\ TFLOPs_{FY30}}$$

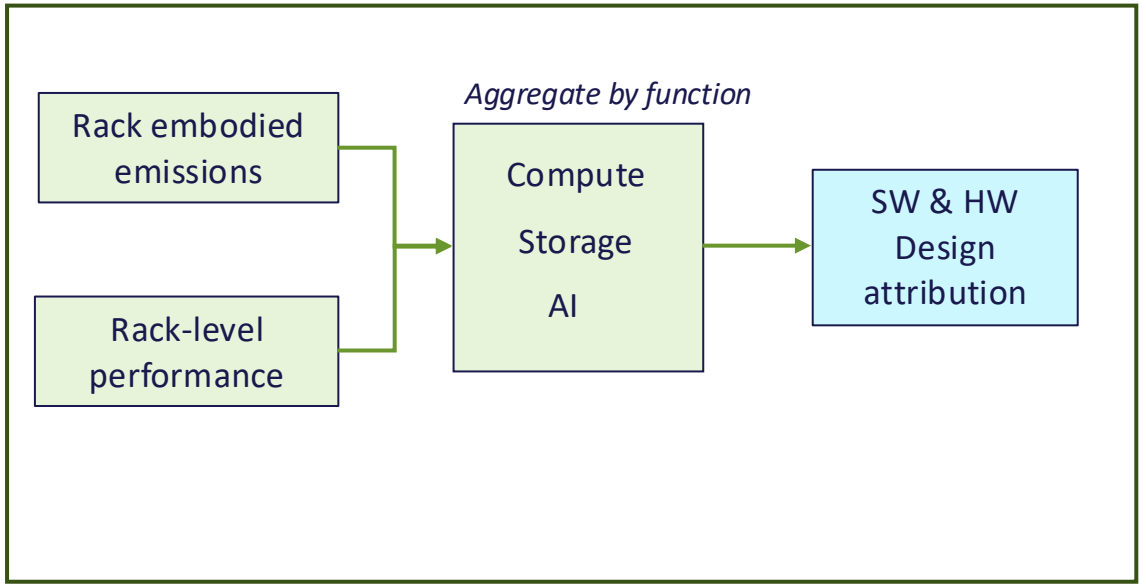
$$\boldsymbol{Avoided\ Emissions_{FY30}} = ReferenceEmissions_{FY30} - BaselineEmissions_{FY30}$$

Multi-level carbon intensity framework: supplier to portfolio

Supplier Emissions Intensity



Portfolio Emissions Intensity



Industry methodology alignment – Taxonomy, functional units, data quality, attribution

- Granular tracking of performance
- Build bottom-up part-level intensity metrics to understand how suppliers have contributed to 'fleet-level' efficiency

- High-level tracking of trends & roadmap-level attribution
- Consolidated by HW function, useful for assessing fleet-wide Scope 3 efficiency
- Does not focus on gen over gen SKU-level improvements

Key Takeaways

- Intensity metrics are useful for communicating progress in the era of growth
- However, absolute emissions must remain top of mind to meet aggressive net-zero targets
- Embodied emissions intensity metrics are not a replacement for full lifecycle metrics needed for comprehensive avoided emissions calculations

Call to Action

- Standardize functional units and calculation methods for compute, storage, and AI hardware.
- Build and aggregate intensity metrics from supplier components through product, rack, and portfolio levels.
- Align supply-chain attribution rules for supplier clean energy, lower-carbon materials, and hardware and software efficiency.
- Pilot the framework across representative systems and refine taxonomy, data quality, comparability, and reporting.

Thank You!

